

MODULE – 5 (Application Layer)

1)Domain Name System (DNS):-

1. Introduction

The **Domain Name System (DNS)** is a system used on the internet to convert human-friendly website names into machine-friendly IP addresses. Since computers understand numbers and not names, DNS acts like a translator.

For example, when a user types **google.com**, DNS converts it into an IP address like **142.250.xxx.xxx**, allowing the browser to connect to the correct server.

2. Working of DNS (Step-by-Step Process)

The DNS resolution process takes place in multiple steps:

1. **User Request:**
The user enters a domain name (e.g., www.geeksforgeeks.org) in the browser.
2. **Local Cache Check:**
The browser or operating system first checks if the IP address is already stored.
3. **DNS Resolver Query:**
If not found, the request is sent to a DNS resolver (usually provided by ISP).
4. **Root Server:**
The resolver contacts the root server, which directs it to the correct TLD server.
5. **TLD Server:**
The Top-Level Domain server (.com, .org, etc.) provides the address of the authoritative server.
6. **Authoritative Server:**
This server gives the actual IP address of the domain.
7. **Final Response:**
The IP address is returned to the browser, and the website is loaded.

3. Structure of DNS (Hierarchy)

DNS is organized in a hierarchical manner:

1. **Root Level (.)**
The topmost level, represented by a dot (.), acts as the starting point.

2. **Top-Level Domain (TLD)**
Includes extensions like .com, .org, .in.
3. **Second-Level Domain**
The main name of the website (e.g., "google" in google.com).
4. **Subdomain**
Additional parts like www, mail, blog.
5. **Hostname**
Identifies specific servers like web1 or mailserver.

4. Types of Domains

DNS classifies domains into:

1. **Generic Domains (gTLDs):**
Used globally (e.g., .com, .org, .net).
2. **Country Code Domains (ccTLDs):**
Represent specific countries (e.g., .in, .us, .uk).
3. **Reverse Domains:**
Used to convert IP address back to domain name using PTR records.

5. Domain Name Server

A Domain Name Server is responsible for storing DNS records and responding to queries.

- Stores records like A, MX, CNAME, etc.
- Provides IP addresses when requested
- Can act as recursive or authoritative server

6. DNS Lookup Types

DNS lookup refers to the process of resolving domain names:

1. **Recursive Query:**
Resolver finds the complete answer for the client.
2. **Iterative Query:**
Server gives the best possible answer or referral.
3. **Non-Recursive Query:**
Answer is already available, so returned immediately.

7. DNS Caching and TTL

DNS caching improves performance by storing previously resolved results.

- **Caching:** Saves IP addresses temporarily
- **TTL (Time-to-Live):**
Defines how long a record stays in cache
Example: TTL = 3600 seconds → valid for 1 hour

This reduces delay and server load.

8. Reverse DNS Lookup

Reverse DNS converts an IP address back to a domain name.

- Uses **in-addr.arpa (IPv4)** and **ip6.arpa (IPv6)**
- Uses **PTR records**

Uses:

- Network troubleshooting
- Email verification (reducing spam)

9. DNS Record Types

DNS records store different types of information:

- **A Record:** Domain → IPv4 address
- **CNAME Record:** Alias to another domain
- **MX Record:** Mail server information

2) Dynamic Domain Name System (DDNS) :-

1. Introduction

The **Dynamic Domain Name System (DDNS)** is an extension of the **Domain Name System (DNS)** that automatically updates the mapping between a **hostname** (like *myhome.dyndns.org*) and its **IP address** whenever the IP changes.

In normal DNS, changes (like a new IP address) must be updated manually in DNS records. But in real-world networks, especially home networks using **dynamic IPs**, IP addresses change frequently. DDNS solves this problem by updating DNS records **automatically in real-time**.

2. Need for DDNS

- ISPs assign **dynamic IP addresses** that change over time
- Manual DNS updates are slow and impractical
- Users need **continuous remote access** (e.g., CCTV, servers)

Therefore, DDNS ensures that a **fixed hostname always points to the latest IP address**.

3. Working of DDNS (Step-by-Step)

Step 1: IP Address Assignment

A device connects to a network and gets an IP using **Dynamic Host Configuration Protocol (DHCP)**.

- Example: IP = 123.45.67.89

Step 2: DDNS Client Detects Change

- A **DDNS client** (in router/software) detects the new IP
- It sends an update request to the DNS server

Step 3: DNS Server Updates Record

- Primary DNS server updates the hostname mapping
- Example:

myhome.dyndns.org → 123.45.67.89

Step 4: Notification to Secondary Servers

- **Active:** Primary informs secondary servers
- **Passive:** Secondary servers periodically check

Step 5: Zone Transfer

- Secondary DNS servers copy updated records
- Ensures consistency across all DNS servers

Step 6: Secure Update

Security methods like:

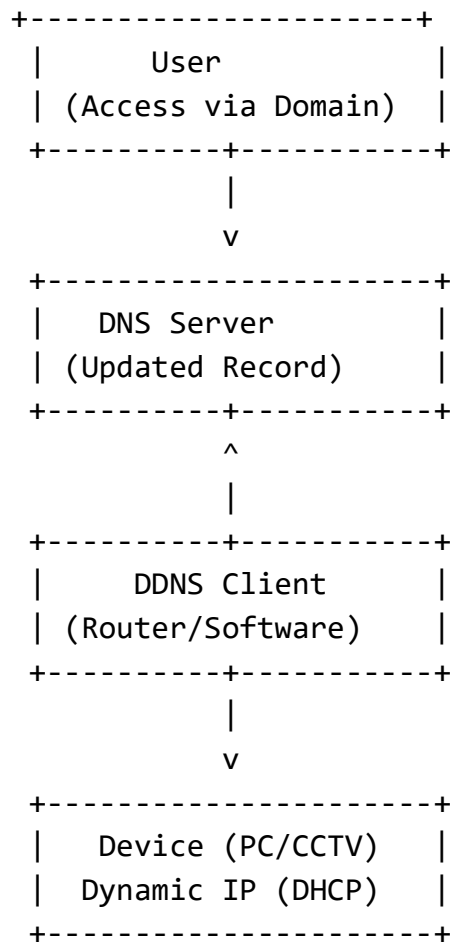
- **Transaction Signature (TSIG)**
- **DNSSEC**

These prevent unauthorized updates.

Step 7: User Access

- User uses hostname instead of IP
- Even if IP changes, hostname still works

4. Diagram of DDNS Working



5. Example

- Initial IP: 123.45.67.89
- Hostname: *myhome.dyndns.org*

Later ISP changes IP → 98.76.54.32

- DDNS updates automatically
- User still uses same hostname

No need to remember new IP

6. Advantages

1. **Easy Remote Access** – Access devices from anywhere
2. **Automatic Updates** – No manual work needed
3. **Cost-effective** – Often free services
4. **Supports Dynamic IPs** – Works well with home networks
5. **Wide Compatibility** – Works with VPN, FTP, IoT

7. Disadvantages

1. **Dependency on Provider** – If DDNS service fails, access stops
2. **Security Risks** – Can expose devices if not secured
3. **Latency Issues** – Delay in updating IP sometimes
4. **Limited Control** – Managed by third-party providers

8. Applications / Use Cases

- Remote access to home computers and CCTV
- Hosting personal websites
- VPN connections
- IoT and home automation systems
- Online gaming servers

3) TELNET:-

TELNET (Teletype Network) is a **client-server protocol** used to remotely access and control another computer over a network (LAN or Internet). It allows a user to log in to a remote system and execute commands as if working directly on that system.

In simple terms:

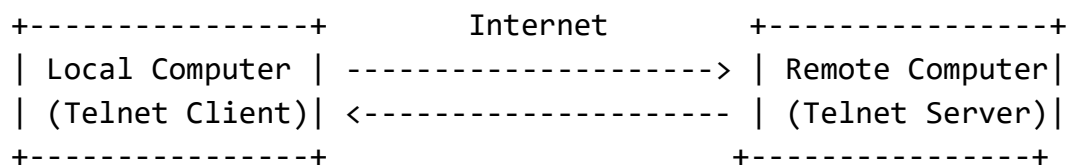
TELNET lets you **use another computer from your own computer.**

1. Basic Working of TELNET

TELNET works on a **client-server model**:

- **Telnet Client** → Runs on the local machine (user side)
- **Telnet Server** → Runs on the remote machine

Diagram: TELNET Communication



The client sends commands → server processes → sends back results.

2. Local Login vs Remote Login

(a) Local Login

User logs into their own system.

Diagram:

User → Keyboard → OS → Application

The system directly processes user input.

(b) Remote Login (TELNET)

User logs into another computer using TELNET.

Diagram:

User → Local OS → Telnet Client → Internet → Telnet Server → Remote OS
→ Application

The remote system processes commands and sends results back.

3. Network Virtual Terminal (NVT)

TELNET uses **Network Virtual Terminal (NVT)** to standardize communication.

Problem: Different computers use different formats

Solution: NVT converts all data into a **common format**

Diagram:

Local Format → NVT Format → Internet → NVT Format → Remote Format

This ensures compatibility between different systems.

4. Step-by-Step Working of TELNET

1. User types command on local computer
2. Telnet client converts it into **NVT format**
3. Data travels through **TCP/IP network**
4. Remote server converts NVT into its own format
5. Command is executed on remote system
6. Output is sent back to the user

5. TELNET Commands

TELNET commands use a special prefix called:

IAC (Interpret As Command) = 255

Important Commands:

Command	Meaning
WILL	Agree to enable option
WON'T	Refuse option
DO	Request/approve option
DON'T	Reject option

These commands control communication between client and server.

6. Modes of Operation

(a) Default Mode

- User types full line
- Sent only after pressing Enter
- Client handles echo

(b) Character Mode

- Each character sent immediately
- Server handles echo

(c) Line Mode

- Editing done on client side
- Entire line sent at once

7. Uses of TELNET

- Remote system administration
- Network troubleshooting
- Learning command-line operations
- Accessing old systems (BBS)

8. Advantages

- Easy remote access
- Works across different systems
- Saves time (no physical presence needed)

9. Disadvantages

- **Not secure** (data sent in plain text)
- Vulnerable to hacking
- Replaced by secure alternatives like SSH

4) Electronic Mail (E-Mail):-

Introduction:

Electronic Mail (E-mail) is a method of sending and receiving digital messages over the Internet using devices like computers and smartphones. It allows users to communicate instantly across the world using a unique email address (e.g., user@example.com). Apart from text, emails can also carry attachments such as documents, images, videos, and audio files.

Key Components of an E-Mail System

An email system consists of several components that work together to ensure smooth communication:

1. **User Agent (UA)**
It is the email application used by users to compose, send, receive, and manage emails.
Examples: Gmail, Outlook.
2. **Mail Server**
A centralized system that stores emails and manages sending and receiving messages.
3. **Message Transfer Agent (MTA)**
It is responsible for transferring emails from the sender's mail server to the receiver's mail server.
4. **Mailbox**
A storage area on the mail server where incoming emails are stored until the user reads them.
5. **Spool File**
A temporary storage area where outgoing emails are kept before they are sent.

Working of E-Mail

The working of email involves multiple steps from sender to receiver:

```
[Sender]
  ↓ (Compose in Email Client)
[User Agent]
  ↓ (SMTP)
```

[Sender Mail Server]
↓ (MTA Routing via DNS/MX)
[Receiver Mail Server]
↓ (Stored in Mailbox)
[Receiver User Agent]
↓ (IMAP/POP3)
[Receiver Reads Email]

Explanation:

- The sender composes an email using a User Agent.
- The email is sent to the sender's mail server using SMTP.
- The Message Transfer Agent (MTA) finds the receiver's server and forwards the message.
- The receiver's mail server stores it in the mailbox.
- The receiver accesses it using IMAP or POP3 and reads it.

Email Protocols

Protocols define how emails are sent and received:

1. SMTP (Simple Mail Transfer Protocol)

- Used for sending emails.
- Works as a push protocol.
- Transfers emails between servers.
- Ports: 25, 587, 465.

2. POP3 (Post Office Protocol Version 3)

- Used to download emails to a device.
- Emails may be deleted from server after download.
- Suitable for single-device access.
- Port: 110 / 995 (secure).

3. IMAP (Internet Message Access Protocol)

- Keeps emails on the server.
- Synchronizes across multiple devices.

- Suitable for modern usage (mobile + laptop).
- Port: 143 / 993 (secure).

Advantages of E-Mail

- **Fast Communication:** Messages are delivered within seconds worldwide.
- **Low Cost:** No need for paper or postage.
- **Supports Attachments:** Files, images, and documents can be sent easily.
- **Accessibility:** Can be accessed from anywhere.
- **Record Keeping:** Emails can be stored and searched.
- **Group Communication:** Send to multiple users using CC/BCC.
- **Improves Productivity:** Useful for formal and business communication.

Limitations of E-Mail

- **Spam Emails:** Unwanted messages clutter inboxes.
- **Phishing Attacks:** Fake emails can steal sensitive data.
- **Malware Risk:** Attachments may contain viruses.
- **Delivery Delay:** Sometimes emails are delayed.
- **Size Limits:** Large attachments may not be allowed.
- **Privacy Issues:** Emails may be intercepted if not secure.
- **Miscommunication:** Tone and intent may be misunderstood.

5)File Transfer Protocol (FTP):-

1. Definition and Purpose

File Transfer Protocol (FTP) is a standard protocol of the TCP/IP suite used to transfer files between computers over a network. It allows a user (client) to access files stored on a remote computer (server) and perform operations like uploading, downloading, and managing files.

In simple terms, FTP works like a bridge that helps computers exchange files even if they use different operating systems.

2. Key Features of FTP

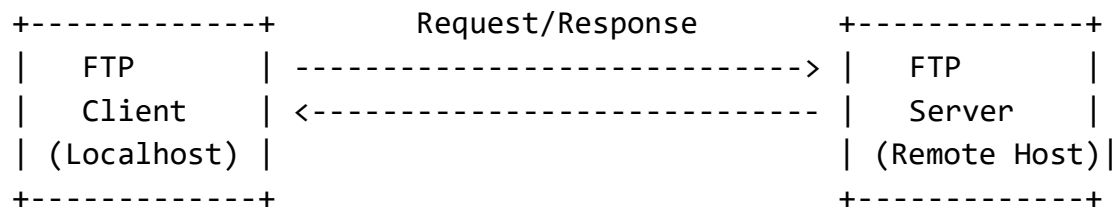
- Enables **file sharing** between different systems.
- Hides differences in **file formats and operating systems**.
- Allows **direct interaction with remote directories**.
- Supports transfer of **text, binary, and large files**.

3. FTP Architecture (Client–Server Model)

FTP works using a **client-server model**:

- **FTP Client**: Software used by the user to request files (e.g., browser, command line).
- **FTP Server**: Stores files and responds to client requests.

Diagram: FTP Client-Server Model



4. Working of FTP

FTP uses **two separate connections**:

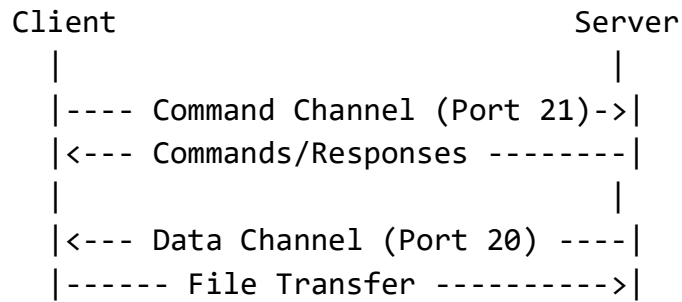
(i) Command Channel

- Used for sending commands (login, file request)
- Works on **Port 21**

(ii) Data Channel

- Used for actual file transfer
- Works on **Port 20**

Diagram: FTP Working



Steps of Operation

1. Client connects to server using Port 21
2. User logs in (username/password or anonymous)
3. Client navigates directories
4. File transfer starts via data channel
5. Data connection closes after transfer
6. Command connection remains active

5. Types of FTP Connections

(A) Active FTP

- Client opens command channel
- Server initiates data connection

Client -----> Server (Command)

Client <----- Server (Data)

Problem: Blocked by firewalls

(B) Passive FTP

- Client opens both command and data connections

Client -----> Server (Command)

Client -----> Server (Data)

Advantage: Works with firewalls (default method)

6. FTP Commands (Important)

- `get` → Download file
- `mget` → Download multiple files
- `ls / dir` → List files
- `cd` → Change directory
- `pwd` → Show current directory
- `quit` → Exit session

7. Transmission Modes in FTP

- **Stream Mode** → Default, continuous data flow
- **Block Mode** → Data sent in blocks with headers
- **Compressed Mode** → Data compressed before transfer

8. Anonymous FTP

- No login required
- Username: *anonymous*
- Password: Email address
- Limited access (download only)

9. Applications of FTP

- Business file sharing
- Backup and recovery systems
- Financial data transfer
- Team collaboration and project sharing

10. Why FTP is Used

- Reliable file transfer mechanism
- Supports multiple file formats (ASCII, Binary, EBCDIC)
- Works across different systems (heterogeneous environment)

6)World Wide Web (WWW): Structure and Functioning:-

1. Introduction

The **Tim Berners-Lee** developed the World Wide Web in 1989 at **CERN** to help researchers share information easily.

The **WWW** is a system of interconnected web pages that are accessed through the Internet using web browsers. It allows users to view text, images, videos, and other multimedia content.

Important Point:

The Web is **a part of the Internet**, not the Internet itself.

2. Structure of WWW (Architecture)

The Web follows a **Client–Server Architecture**, which includes:

(a) Client (User Side)

- Devices like computers or mobile phones
- Uses web browsers such as **Google Chrome**
- Sends requests to servers

(b) Server

- A powerful computer that stores websites and data
- Responds to client requests

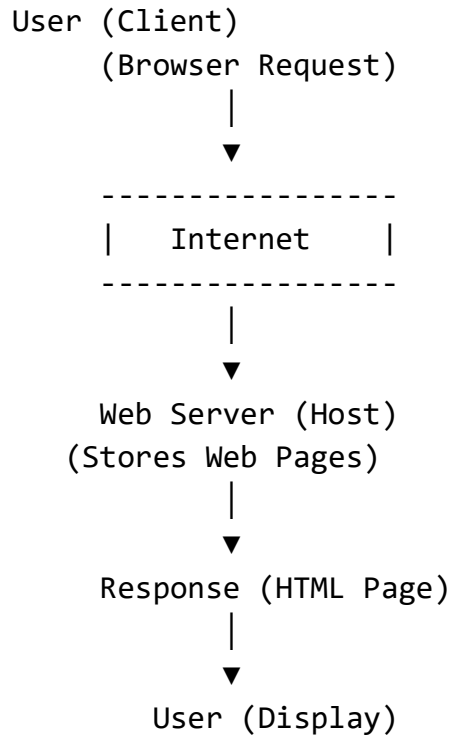
(c) Communication Protocols

- **HTTP (Hypertext Transfer Protocol)** → Transfers web data
- **HTTPS** → Secure version of HTTP

(d) Web Resources

- Web pages written in **HTML**
- Identified using **URL (Uniform Resource Locator)**

3. Diagram of WWW Structure



This diagram shows how the client communicates with the server through the Internet.

4. Functioning of WWW (Working Process)

The working of the Web can be explained step-by-step:

Step 1: URL Request

- The user enters a URL in the browser
Example: www.example.com

Step 2: DNS Resolution

- The domain name is converted into an IP address

Step 3: Request to Server

- The browser sends an HTTP request to the web server

Step 4: Server Processing

- The server finds the requested webpage

Step 5: Response

- Server sends back the webpage (HTML, CSS, JS files)

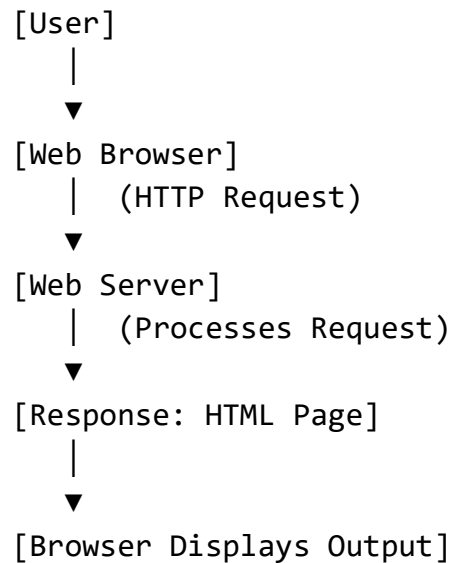
Step 6: Rendering

- Browser displays the page to the user

5. Diagram of WWW Functioning

User → Browser → HTTP Request → Web Server
User ← Browser ← HTTP Response ← Web Server

Or more detailed:



6. Key Components Involved

- **URL** → Address of webpage
- **HTTP/HTTPS** → Communication rules
- **HTML** → Structure of webpage
- **Web Browser** → Interface for users

7) HTTP methods and HTTP headers:

1. HTTP Methods :-

HTTP Methods are **standard actions (operations)** that a client performs on a server to request or manipulate resources (like web pages, data, etc.). They define **what kind of request is being made**.

Common HTTP Methods with Explanation & Example

1. GET

- Used to **retrieve data** from the server
- Does **not modify** anything
- Example: Opening a website

Example:

GET /home HTTP/1.1

Real-life: Viewing a YouTube page

2. POST

- Used to **send data** to the server
- Creates a new resource

Example:

POST /register HTTP/1.1

Real-life: Submitting a signup form

3. PUT

- Used to **update an existing resource completely**

Example:

PUT /user/101 HTTP/1.1

Real-life: Updating full profile info

4. PATCH

- Used to **partially update data**

Example:

PATCH /user/101

Real-life: Updating only email

5. DELETE

- Used to **remove a resource**

Example:

DELETE /user/101

Real-life: Deleting an account

Diagram: HTTP Methods Flow

Client (Browser)

|
| GET /data
v

Server -----> Sends Data

Client

|
| POST /data
v

Server -----> Stores Data

Key Points (Exam Writing)

- Methods define **type of operation**
- Used in **REST APIs and web applications**
- Some are **safe (GET)** and some **modify data (POST, PUT, DELETE)**

2. HTTP Headers :-

HTTP Headers are **key-value pairs** that carry **extra information** between client and server during request and response.

They help in:

- Communication
- Authentication
- Content type identification
- Caching

Types of HTTP Headers

1. Request Headers

- Sent by client to server

Example:

Host: example.com
User-Agent: Chrome
Authorization: Bearer token

2. Response Headers

- Sent by server to client

Example:

Content-Type: text/html
Server: Apache

3. General Headers

- Used in both request and response

Example:

Cache-Control: no-cache
Connection: keep-alive

4. Entity Headers

- Describe body/content

Example:

Content-Length: 348

Content-Encoding: gzip

Diagram: HTTP Headers in Communication

Client Request

```
-----  
GET /index.html  
Host: example.com  
User-Agent: Chrome  
-----
```

↓

Server Response

```
-----  
HTTP/1.1 200 OK  
Content-Type: text/html  
Content-Length: 1024  
-----
```

Key Points

- Headers = **metadata (information about data)**
- Sent in **both request & response**
- Important for **security, caching, content handling**

8) Simple Network Management Protocol (SNMP):-

Introduction

Simple Network Management Protocol (SNMP) is a standard protocol used for **monitoring and managing network devices** such as routers, switches, servers, printers, firewalls, and load balancers connected through an IP network.

It works at the **Application Layer** of the TCP/IP model and mainly uses **UDP ports 161 and 162** for communication.

The main purpose of SNMP is to help network administrators:

- Monitor network performance
- Detect network faults
- Troubleshoot problems
- Configure remote devices
- Plan network expansion

SNMP provides a **common method** to collect information from devices made by different manufacturers.

1. SNMP Architecture

SNMP architecture consists of **three main components**:

(a) SNMP Manager (Network Management Station - NMS)

The **SNMP Manager** is the central system that controls and monitors all devices in the network.

Functions:

- Sends requests to devices
- Collects information
- Displays performance reports
- Detects errors and faults

Example:

A network monitoring server checking CPU usage of all routers.

(b) SNMP Agent

The **SNMP Agent** is software installed on managed devices like routers or switches.

It collects device information and sends it to the manager.

Functions:

- Stores management data
- Responds to manager requests

- Sends alerts when problems occur

Example:

A router agent reporting packet loss to the manager.

(c) Management Information Base (MIB)

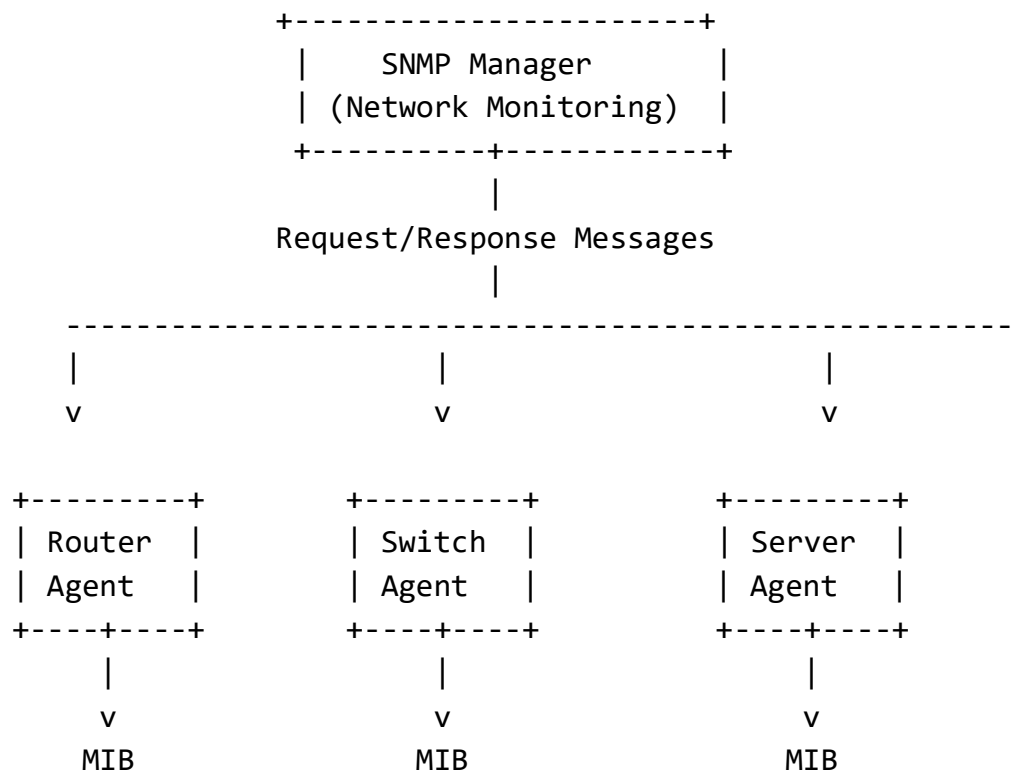
The **MIB** is a database that stores information about managed devices.

It contains variables called **managed objects** such as:

- CPU usage
- Memory usage
- Interface traffic
- Error count

These objects are arranged in a **hierarchical tree structure**.

Diagram of SNMP Architecture



2. Working of SNMP

SNMP works on a **manager-agent communication model**.

Step-by-step Working:

Step 1:

The SNMP manager sends a request to the agent.

Step 2:

The agent checks the requested data in its MIB.

Step 3:

The agent sends the response back.

Step 4:

If any fault occurs, the agent sends a trap message automatically.

Example

Suppose a manager wants to know router traffic.

- Manager sends request
- Router agent checks traffic data
- Agent replies with traffic statistics

If router crashes:

- Agent sends trap message immediately

3. SNMP Message Types

SNMP uses different messages for communication.

(a) GetRequest

Used by manager to retrieve information from an agent.

Example:

Checking CPU utilization of a switch.

(b) GetNextRequest

Used to retrieve the next value in a MIB table.

Useful when the manager does not know exact object index.

Example:

Reading all interface details one by one.

(c) SetRequest

Used to change the value of an object in the device.

Example:

Changing device configuration remotely.

(d) Response

Sent by the agent as a reply to Get or Set requests.

(e) Trap

Sent automatically by the agent when an event or fault occurs.

Example:

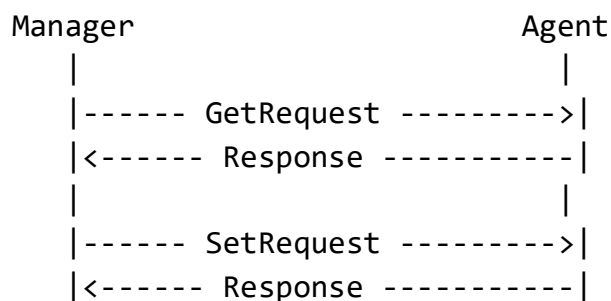
Link failure alert

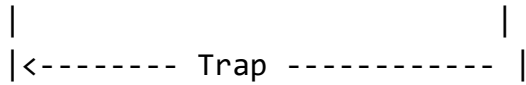
(f) InformRequest

Similar to trap, but requires acknowledgement from manager.

Ensures that manager has received the message.

SNMP Message Flow Diagram





4. SNMP Security Levels

SNMP provides three security levels.

(a) noAuthNoPriv

- No authentication
- No encryption

Least secure.

(b) authNoPriv

- Authentication provided
- No encryption

Moderate security.

(c) authPriv

- Authentication
- Encryption

Most secure.

Used in sensitive networks.

5. Versions of SNMP

(a) SNMPv1

First version.

Features:

- Uses community strings
- Simple setup
- Low security

(b) SNMPv2c

Improved version of SNMPv1.

Features:

- Better performance
- More message types
- Still uses community strings

(c) SNMPv3

Most secure version.

Features:

- Authentication
- Encryption
- Better access control

This is the **most widely preferred version today**.

Version Comparison Table

Version	Security	Authentication	Encryption
SNMPv1	Low	Community String	No
SNMPv2c	Medium	Community String	No
SNMPv3	High	MD5/SHA	DES/AES

6. Characteristics of SNMP

SNMP has the following characteristics:

- Used for network monitoring
- Detects faults quickly
- Supports remote configuration
- Standardized protocol
- Works with multi-vendor devices

7. Advantages of SNMP

Easy to Implement

Simple setup and configuration.

Low Resource Usage

Consumes minimal CPU and memory.

Vendor Independent

Works across devices from different manufacturers.

Fault Detection

Quickly detects network problems.

Extensible

Can support new devices and features.

8. Limitations of SNMP

Poor Scalability

Performance reduces in very large networks.

High Polling Overhead

Frequent polling increases traffic.

Limited Data Representation

Not object-oriented.

Security Issues in Older Versions

SNMPv1 and v2c are less secure.

Vendor-Specific Extensions

Different implementations may create compatibility issues.

9. Applications of SNMP

SNMP is used for:

- Monitoring bandwidth usage
- Checking server health
- Detecting link failures
- Monitoring printer status
- Managing switches and routers

9)Bluetooth:

1. Definition

Bluetooth is a **short-range wireless communication technology** that allows devices to exchange data using **radio waves instead of cables**. It is mainly used for connecting nearby devices like phones, laptops, and accessories.

It works in the **2.4 GHz ISM band** and is designed for **low power and personal use**.

2. Types of Bluetooth

(a) Bluetooth Classic (BR/EDR)

- Used for **continuous data transfer** like music or calls
- Higher power consumption
- Examples: headphones, speakers

(b) Bluetooth Low Energy (BLE)

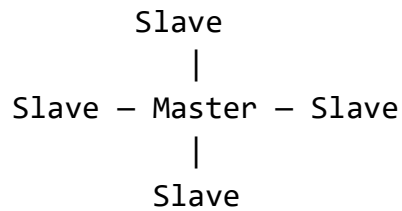
- Used for **small, occasional data transfer**
- Very low power consumption
- Examples: smartwatches, fitness bands

3. Architecture of Bluetooth

(a) Piconet

A small Bluetooth network where:

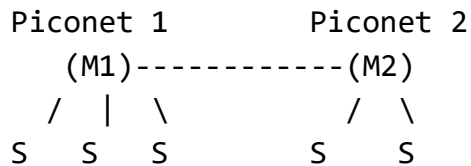
- One device = **Master**
- Up to 7 devices = **Slaves**



Master controls communication and timing.

(b) Scatternet

Multiple piconets connected together.



A device can act as:

- Master in one network
- Slave in another

4. Bluetooth Protocol Stack (Layers)

This defines how Bluetooth communication happens.

Application Layer (Profiles)

RFCOMM | SDP | OBEX

L2CAP (Data handling)

LMP (Link management)

Baseband (Connection control)

Radio Layer (Transmission)

Important Layers Explained

- **Radio Layer** → Sends signals through air
- **Baseband** → Controls connections
- **LMP** → Security (authentication, encryption)
- **L2CAP** → Data transfer management
- **SDP** → Finds services on other devices
- **RFCOMM** → Acts like a cable replacement

5. Applications of Bluetooth

- Wireless headphones, keyboards, speakers
- File transfer between phones/laptops
- Smart devices (IoT, fitness bands)
- Healthcare monitoring devices
- Car hands-free systems

6. Advantages

- No cables required (wireless)
- Low power consumption
- Easy to use and low cost
- Supports voice and data

7. Disadvantages

- Limited range (around 10–100 meters)
- Slower than Wi-Fi
- Can face interference (same 2.4 GHz band)
- Security risks if not properly configured

10)Firewall in Computer Networks:-

1. Introduction

A **firewall** is a security system (hardware or software) that monitors and controls network traffic based on predefined rules. It acts as a **barrier between a trusted internal network (LAN)** and an **untrusted external network (WAN/Internet)**.

In simple terms, a firewall works like a **security guard at a gate**:

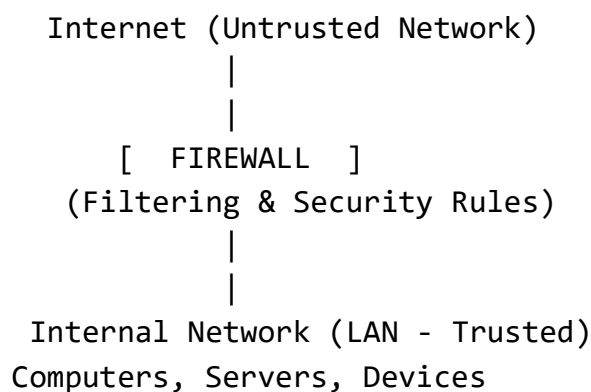
- It allows safe traffic
- blocks harmful traffic
- and decides what can enter or leave the network

2. Basic Firewall Actions

A firewall can take three main actions on data packets:

- **Accept** → Allows the data to pass
- **Reject** → Blocks the data and sends an error message
- **Drop** → Silently blocks the data (no response)

3. Diagram of Firewall



All incoming and outgoing data must pass through the firewall.

4. Working of Firewall

The firewall works in the following steps:

1. **Packet Arrival**

Data enters or leaves the network in the form of packets.

2. **Inspection**

The firewall checks each packet (IP address, port number, protocol, etc.).

3. **Rule Matching**

The packet is compared with predefined security rules.

4. **Decision Making**

- a. Safe → Allowed
- b. Suspicious → Blocked

5. **Logging & Alerts**

Blocked or unusual activities are recorded and alerts may be generated.

6. **Default Policy**

If no rule matches:

- a. Firewall applies default action (Accept / Reject / Drop)
- b. Best practice → **Drop or Reject**

5. **Importance of Firewall**

A firewall is essential because it:

- **Prevents Unauthorized Access**
Only trusted users can access the network
- **Blocks Malicious Traffic**
Stops viruses, malware, phishing attacks, and DoS attacks
- **Protects Sensitive Data**
Prevents data theft and leakage
- **Controls Network Usage**
Applies restrictions (e.g., office policies, parental control)
- **Reduces Insider Threats**
Detects suspicious activities inside the network

6. **Types of Firewall**

(A) **Based on Network Placement**

- **Packet Filtering Firewall** → Checks packets based on IP and ports
- **Stateful Inspection Firewall** → Tracks active connections
- **Proxy Firewall (Application Level)** → Acts as an intermediary
- **Circuit-Level Gateway** → Monitors session connections
- **Web Application Firewall (WAF)** → Protects web applications

- **Next-Generation Firewall (NGFW)** → Advanced security with deep inspection

(B) Based on Systems Protected

- **Network Firewall** → Protects entire network
- **Host-Based Firewall** → Protects individual device

(C) Based on Data Filtering Method

- **Perimeter Firewall** → At network boundary
- **Internal Firewall** → Inside network
- **Distributed Firewall** → Spread across multiple systems

(D) Based on Form

- **Hardware Firewall** → Physical device
- **Software Firewall** → Installed program

11)Cryptography:

1. Definition of Cryptography

Cryptography is the technique of securing information by converting readable data (**plaintext**) into an unreadable form (**ciphertext**) so that only authorized users can access it.

It ensures safe communication over insecure networks like the Internet.

2. Basic Working of Cryptography

Cryptography works in two main steps:

- **Encryption** → Converts plaintext into ciphertext
- **Decryption** → Converts ciphertext back into plaintext

Diagram: Basic Cryptography Process

Plaintext —(Encryption + Key)→ Ciphertext —(Decryption + Key)→ Plaintext
 (Message) (Encrypted Data)
 (Original Message)

3. Goals of Cryptography

Cryptography ensures the following security goals:

1. **Confidentiality** – Data remains secret
2. **Integrity** – Data is not altered
3. **Authentication** – Sender identity is verified
4. **Non-repudiation** – Sender cannot deny sending the message

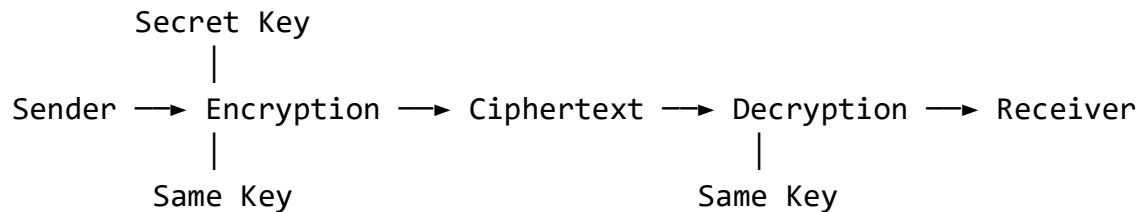
4. Types of Cryptography

(a) Symmetric Key Cryptography

- Same key is used for encryption and decryption
- Fast and efficient

Example: AES, DES

Diagram: Symmetric Encryption



(b) Asymmetric Key Cryptography

- Uses two keys:
 - Public Key (for encryption)
 - Private Key (for decryption)

Example: RSA

Diagram: Asymmetric Encryption



(c) Hash Functions

- Converts data into fixed-length hash value

- Used for passwords and data integrity

Example: SHA, MD5

Diagram: Hashing

Message → Hash Function → Fixed Length Hash

5. Classical Cryptography Techniques

Used in old systems (before computers):

(a) Substitution Cipher

- Letters are replaced by other letters

Example: Caesar Cipher

A → D, B → E, C → F

HELLO → KHOOR

(b) Transposition Cipher

- Letters are rearranged, not replaced

Example: Rail Fence Cipher

HELLO → HLOEL

6. Modern Cryptography Algorithms

(a) DES (Data Encryption Standard)

- Uses 56-bit key
- Now considered insecure

(b) AES (Advanced Encryption Standard)

- Strong and widely used
- Used in banking, HTTPS

7. Applications of Cryptography

Cryptography is used in:

- **HTTPS websites**
- **Online banking & payments**
- **Digital signatures**
- **Password protection**
- **Secure messaging (WhatsApp, Email)**

8. Cryptanalysis (Attacks on Cryptography)

These are methods used to break encryption:

- **Brute Force Attack** – Trying all keys
- **Dictionary Attack** – Using common passwords
- **Rainbow Table Attack** – Precomputed hashes

9. Advantages of Cryptography

- Protects sensitive data
- Ensures secure communication
- Builds trust in digital systems

10. Disadvantages of Cryptography

- Complex to implement
- Key management is difficult
- If key is lost → data cannot be recovered